Project Title

Fine-Tuning DistilGPT-2 using LoRA for an AI Instruction Assistant

GitHub Repository | Google collab Notebook

1. Introduction

Large Language Models (LLMs) have revolutionized Natural Language Processing (NLP) by enabling machines to generate human-like text. Although pre-trained models perform well on general tasks, they often require fine-tuning for instruction-specific applications.

In this project, **DistilGPT-2** was fine-tuned using the **LoRA (Low-Rank Adaptation)** technique on the **yahma/alpaca-cleaned** dataset. The objective was to build an AI Instruction Assistant capable of generating relevant and meaningful responses while keeping training computationally efficient.

2. Use Case

The project aims to develop an **AI Instruction Assistant** capable of understanding user instructions and generating relevant responses.

Applications include:

- Answering general knowledge questions
- Explaining technical concepts
- Providing productivity suggestions
- Generating motivational quotes
- Assisting with simple instruction-based tasks

This demonstrates how parameter-efficient fine-tuning can adapt a pre-trained language model for real-world conversational applications.

3. Dataset

The project uses the **yahma/alpaca-cleaned** dataset, a cleaned version of the Stanford Alpaca dataset designed for instruction-following tasks.

Each record contains:

- **Instruction**
- **Input (Optional)**
- **Output**

Only a subset of the dataset was used for training to reduce training time while maintaining good performance.

Dataset Statistics

Parameter	Value
Dataset	yahma/alpaca-cleaned
Total Samples	51,760
Training Samples	10,000
Validation Samples	2,000

Example Dataset Record

Instruction

Explain Machine Learning.

```
ask("Explain machine learning.")

Instruction: Explain machine learning.
Answer: Machine Learning is a mathematical concept that has been used to improve the performance of computer programs and applications by increasing efficiency in processing data, rec
```

4. Model Used

DistilGPT-2

DistilGPT-2 is a lightweight version of GPT-2 that provides efficient text generation with fewer parameters. Its smaller size makes it suitable for fine-tuning on Google Colab while maintaining good language generation performance.

Model Details

Parameter	Value
Model	DistilGPT-2
Architecture	Decoder-only Transformer
Parameters	~82 million
Framework	Hugging Face Transformers
Task	Causal Language Modelling

5. Technique Used

Low-Rank Adaptation (LoRA)

LoRA is a Parameter-Efficient Fine-Tuning (PEFT) technique that updates only a small number of trainable parameters while keeping the original model weights frozen. This reduces memory usage and training time without significantly affecting model performance.

LoRA Configuration

Parameter	Value
Rank	16
Alpha	32
Dropout	0.1
Target Module	c_attn

Benefits

- Reduced GPU memory usage
- Faster training
- Efficient fine-tuning
- Lower computational cost

6. Training Process

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7. Flow Diagram

```
        Alpaca Dataset
              │
              ▼
      Data Preprocessing
              │
              ▼
         Tokenization
              │
              ▼
       Load DistilGPT-2
              │
              ▼
      Apply LoRA Adapters
              │
              ▼
    Fine-Tune using Trainer
              │
              ▼
     Save Fine-Tuned Model
              │
              ▼
      Generate Responses
```